# Math142 - Practice Problems for Exam 2

1. Determine whether the sequence is convergent or divergent. If it is convergent, find its limit.
(a) $a_n = \frac{n \sin n}{n^2+1}$
(b) $a_n = (1 + \frac{3}{n})^n$

2. Determine whether the series is convergent or divergent.
(a) $\sum_{n=1}^{\infty} \ln(\frac{n}{3n+1})$
(b) $\sum_{n=1}^{\infty} \frac{n}{\sqrt{n^5+n+1}}$
(c) $\sum_{n=2}^{\infty} \frac{1}{n\sqrt{\ln n}}$
(d) $\sum_{n=1}^{\infty} \frac{n^{2n}}{(1+2n^2)^n}$
(e) $\sum_{n=1}^{\infty} \frac{\cos^2 n}{n^{4/3}}$
(f) $\sum_{n=1}^{\infty} ne^{-n}$
(g) $\sum_{n=1}^{\infty} \frac{3^n n^2}{n!}$
(h) $\sum_{n=1}^{\infty} (-1)^{n-1} \frac{n^n}{n!}$

3. Determine whether the series is conditionally convergent, absolutely convergent or divergent.
(a) $\sum_{n=1}^{\infty} (-1)^{n-1} \frac{1}{n^3}$
(b) $\sum_{n=1}^{\infty} (-1)^n \sin(\frac{\pi}{n})$
(c) $\sum_{n=1}^{\infty} \frac{\sin 2n}{1+2^n}$
(d) $\sum_{n=2}^{\infty} (-1)^n \frac{\ln n}{n}$

4. Find the radius of convergence and interval of convergence of the series.
(a) $\sum_{n=1}^{\infty} \sqrt{n} x^n$
(b) $\sum_{n=1}^{\infty} (-1)^n \frac{x^n}{n^2 2^n}$
(c) $\sum_{n=1}^{\infty} \frac{3^n (x-3)^n}{\sqrt{n+3}}$

5. Find the Maclaurin series (power series centered at 0) and its radius of convergence.
(a) $f(x) = \frac{x^2}{1+x}$
(b) $f(x) = xe^{2x}$
(c) $f(x) = \sin(x^2)$.